

- The height of the raised end of the track above the bench is h , as shown in Fig. 2.3.

Measure and record h .

$$h = \dots\dots\dots [1]$$

- (ii) Estimate the percentage uncertainty in your value of h . Show your working.

$$\text{percentage uncertainty} = \dots\dots\dots \% [1]$$

- (c) (i) • Measure and record the diameter d of the smaller sphere.

$$d = \dots\dots\dots$$

- Calculate $1 - \left(\frac{x}{d}\right)^2$.

$$1 - \left(\frac{x}{d}\right)^2 = \dots\dots\dots [1]$$

- (ii) • Place the smaller sphere on the track at the raised end.
- Release the sphere.
 - The time taken for the sphere to roll along the track into the box is t .

Take measurements to determine t .

$$t = \dots\dots\dots [2]$$

